Social Prescribing to promote and improve access to health and care services for people in vulnerable situations in Europe

SP-EU

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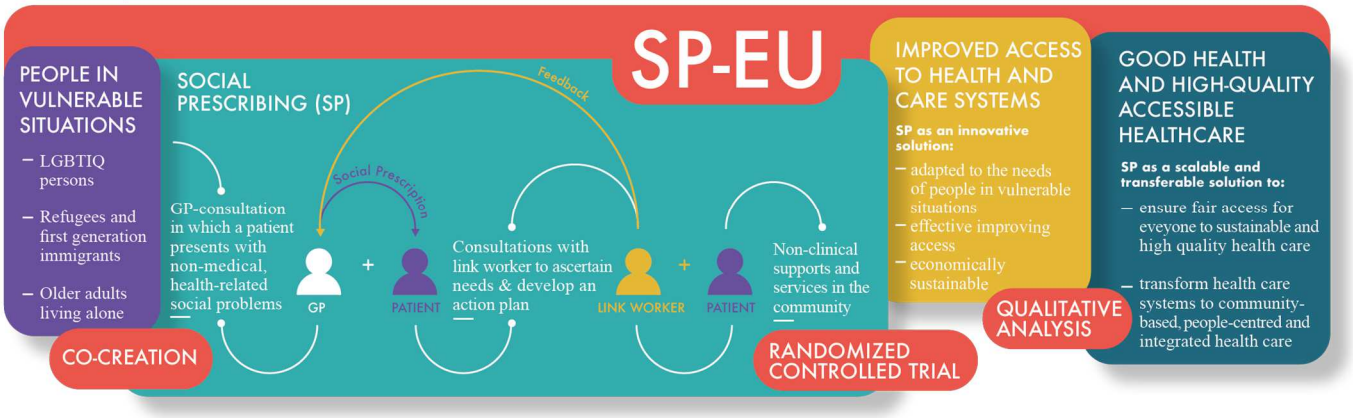
1 Excellence

1.1 Objectives and ambition #@PRJ-OBJ-PO@#



People in vulnerable situations have a lower chance to achieve good health. A major reason for these unequally distributed chances for good health are health-related social problems, such as loneliness, housing problems, financial problems or family difficulties. **Social problems are determinants of health** and can increase the risk for diseases, lower quality of life and impair health as a state of complete physical, mental and social well-being.

In the European Union, there are many **non-clinical supports and services within the community available** provided by the state, NGOs and voluntary organizations to cover these unmet social needs. These services belong to the *"Health and care systems"* with the broader notion *encompassing all parts of health systems and health related parts of social care systems* as described in the call. However, these offers usually have no direct connection to medical care and are often underutilized by people in vulnerable situations. The **first point of contact** for people with social problems is regularly the **general practitioner**'s (GP) office, because social problems often manifest themselves somatically or people simply do not know where to go. However, due to this dis-integration of health and care services, widely existing third sector offers in the community are not adequately accessed.

Social Prescribing (SP) is an innovative solution to bridge this gap, improve access to health and care services and strengthen the community orientation and health promoting aspects of health care. Hence, **SP mitigates the effects of social determinants of health** on health outcomes. SP enables GPs who identify health-related social problems to refer patients to a so-called link worker. Subsequently, these patients have several appointments with a link worker. The link worker provides personalised support and focuses on what matters to the person. Together with the patient, they create an action plan and help them access community resources by addressing any barriers that may exist. SP is one approach out of a wider range of measures to integrate health and social care and strengthen community orientation and health promotion.

There is some **evidence that SP has positive impacts**. Subsequently, SP has been rolled out nationwide in the UK. Several European countries have taken up the idea of SP to differing extents, from no uptake to planning to nationwide rollouts. However, **SP has not been tailored to the needs of people in vulnerable situations** and **the effectiveness of SP to improve access to health and care services for people in vulnerable situations remains unknown**. Thus, it is of utmost importance to tailor SP specifically to people in vulnerable situations, prove the effectiveness of SP for people in vulnerable situations (those who may benefit the most from this intervention) and to provide evidence for the sustainable scalable implementation of SP.

Objectives

Our **vision** is that **SP improves access to health and care services for people in vulnerable situations** and thus **contributes as a scalable and transferable innovative solution to the transformation of health care systems towards community-based, people-centred and integrated health care**.

The **overall objective** of SP-EU is to **assess the potential of SP to promote and improve access to health and care services** (in a broader notion) **for people in vulnerable situations**, focusing on three primary target groups: a) LGBTIQ persons, b) refugees and first-generation immigrants and c) older adults living alone. Our ambition is defined by five major objectives:

Objective 1 is to **involve people in vulnerable situations** belonging to these three groups (a) LGBTIQ persons, b) refugees and first-generation immigrants and c) older adults living alone) in the design and implementation of SP adapted to their needs and social context.

KPI[1] 1.1a,b,c Three co-creation workshops with each group a,b,c (total 9 workshops) until project month (PM) 12
KPI 1.2 A web-based open-source technical solution to support adapted SP until PM 18
KPI 1.3 Publish a manual on co-creation in European health care interventions until PM 54
KPI 1.4a,b,c One co-creative workshop to interpret the results of SP-EU with each group a,b,c until PM 60

Objective 2 is to **assess the effectiveness of SP to improve access to health related parts of social care systems** compared to care as usual for a) LGBTIQ persons, b) refugees and first-generation immigrants and c) older adults living alone by a pragmatic international multicentre randomized controlled basket trial (RCT)

KPI 2.1 Recruit 100 GP practices as study sites in Germany, Switzerland, Denmark, Poland, Spain, Portugal, Belgium, and Croatia until PM 20
KPI 2.2a,b,c Randomize at least 592 primary care patients per group from eight European countries in an RCT on the effectiveness of SP until PM 24 (total 1,776 patients)
KPI 2.3a,b,c Complete a 6-month follow up assessing the primary endpoint (utilization of non-clinical supports and services within the community) of at least 398 participants until PM 30 (total 1,194 patients)
KPI 2.4a,b,c Publish one article on the primary endpoint as a preprint until PM 54
KPI 2.5 Publish one article on the meta-analysis to pool the data of all three sub-trials as a preprint until PM 55
KPI 2.6 Publish one article on the health economic analysis as a preprint until PM 57
KPI 2.7 Publish anonymized trial data until PM 58

Objective 3 is to **explore enabling and limiting factors to implement SP** as a solution to improve access to health and care services for people in vulnerable situations as a scalable and transferable solution for different European countries.

KPI 3.1 Conduct 100 qualitative interviews with stakeholders in the United Kingdom, Poland, Portugal, Austria and Germany until PM 36
KPI 3.2 Publish one article that compares and contrasts qualitative data from across these countries using framework analysis to assess enabling and limiting factors as a preprint until PM 48

Objective 4 to **engage decision- and policy makers, providers, and health and care workers** to appreciate the potential impact of SP and to exploit the project's findings to improve good health and high-quality accessible healthcare.

KPI 4.1a,b Develop an initial exploitation plan until PM 36 and an update exploitation plan until PM 60
KPI 4.2 Conduct one policy roundtable in eight EU countries for decision- and policy makers, providers, health and care workers on how to sustainably implement SP for people in vulnerable situations until PM54
KPI 4.3 Conduct one EU-level roundtable (e.g., in the European Parliament) for decision- and policy makers on SP until PM60
KPI 4.4 Develop a framework for sustainable financing of SP in European countries until PM 60

Objective 5 is to **inform the public - especially people in vulnerable situations - about the potential of SP** and the results of our study on an ongoing basis.

KPI 5.1 Communication via social media channels with at least 600 followers to discuss our project with the public until PM 60
KPI 5.2 Webpage including a knowledge hub with at least 10,000 visits per year until PM 36
KPI 5.3 Communication via classic mass media in at least six countries will mention our project at least twice over the five-year project timeline, e.g., newspapers or breakfast television

Progress beyond the state-of-the-art and ambition

Our ambition is **to establish a scientific base for SP as an innovative solution to promote and improve access to health, social and care services for persons in vulnerable situations**, particularly the target groups identified for this project.

While there is a growing body of evidence on the positive effects of SP in the general population, there is a lack of rigorously conducted RCTs to prove its effectiveness. There has also been no multinational trial to prove effectiveness of SP so far with most studies to date conducted in the UK. SP-EU would conduct the **first multinational randomized-controlled trial on SP**, with the potential for ground-breaking impacts on researching a promising care intervention across different European primary care settings. If funded, this would be **one of the largest European RCTs on health care interventions at the primary care level** ever conducted. By harnessing the basket design, which has so far been mainly used for personalized therapy in oncology, we would like to contribute to **the methodological advancement of trials in health services research**.

The **international co-creation process** will be highly innovative in adapting existing co-creation processes to multinational studies and thus exemplary for scholars and policy makers.

SP shows potential but to maximize any benefits for those most in need stakeholders in policy, practice and academia must work collaboratively to agree evidence needs and communicate best practice to all stakeholders using timely,

[1] Key Performance Indicator

concise outputs. SP-EU will contribute to widening the existing state of the art looking at **implementation processes in different European health care systems** comparatively. The project will carefully look for opportunities and approaches where SP can be made implementable across as many regions of the European Union as possible.

Positioning of the project on the innovation chain ('from lab to market' or 'from idea to application')

The project starts at **Technology Readiness Level 6** (technology demonstrated in relevant environment). The feasibility and positive effects of SP in the primary care setting and the community, as well as for the population in general have been shown. However, there is a lack of evidence on the feasibility and effectiveness for people in vulnerable situations, as it is known that especially these three groups are underserved in current SP programs. In addition, there is only very limited evidence for the effectiveness of SP outside the UK. With this project including co-creation, pragmatic RCT and qualitative implementation research, **we will bring SP to Technology Readiness Level 9** (actual system proven in operational environment) by making SP for people in vulnerable situations widely available in Europe with the necessary evidence to convince relevant policy makers and funders to integrate the concept into usual care.

#§PRJ-OBJ-PO§#

1.2 Methodology #@CON-MET-CM@# #@COM-PLE-CP@#

Overall methodology including underlying concepts, models and assumptions

To deliver our project's objectives, we have chosen a methodological concept, which consists of three major parts: **Co-creation**, a **pragmatic randomized controlled trial (RCT)** and a **qualitative comparative approach** accompanied by **measures regarding communication, dissemination and policy**.

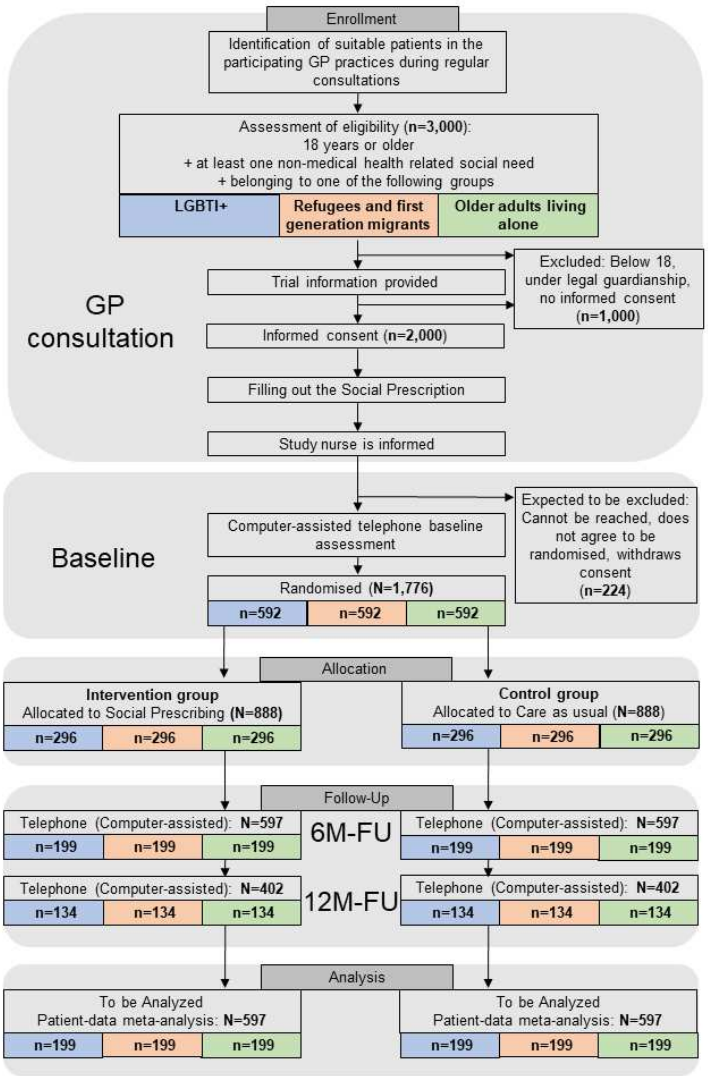
Co-Creation and the RCT will be conducted in ten study hubs in eight different European countries (Belgium, Croatia, Denmark, Germany, Poland, Portugal, Spain, Switzerland). The qualitative analysis will be conducted in five different European countries (Austria, Germany, Poland, Portugal, UK). Communication, dissemination and policy will be supported by an implementation board, which involves representatives from at least six additional countries. All planned activities have been discussed with representatives of the vulnerable groups and external methodological experts during the preparation of the proposal. Details will be further defined during the second stage of the call with all relevant stakeholders to ensure that the perspectives of the target groups are taken into account.

In the **co-creation** part, we will work together with representatives from each group of persons in vulnerable situations (i.e., a) LGBTIQ persons, b) refugees and first-generation immigrants and c) older adults living alone) to adapt SP to their needs and social contexts. Out of the ten study hubs, three pioneer study hubs will focus on one of the vulnerable groups each, and the results from the pioneer hubs will be shared to inform all ten study hubs. Following the framework proposed by Hawkins et al.[2], we will conduct an evidence review and consult key stakeholders along with representatives of each vulnerable group in the 1st phase of the co-creation process. In the 2nd phase, we will co-create an adaptation of SP. In the 3rd phase, we will prototype the intervention with people of each group. This adapted intervention will be the basis for the following pragmatic RCT.

The **pragmatic RCT** will assess the effectiveness of adapted SP for a) LGBTIQ persons, b) refugees and first-generation immigrants and c) older adults living alone, each with non-medical health related social problems. We will conduct a so called 'basket trial', which consists of three independent sub-trials, one for each participant group. We will analyse the results of these sub-trials independently, followed by a pooled analysis through a prospective individual patient data (IPD) meta-analysis. The trial will be a multinational, multicentre, randomized controlled open trial including 10 study hubs, each encompassing about 10 study centres (i.e. GP practices), in 8 different European countries. All study hubs will participate in all sub-trials. For each sub-trial, we will include about 600 patients from the 100 GP practices into the study. The primary outcome of the study is 'access to health related parts of social care systems', measured by the utilization of non-clinical supports and services within the community[3] within 6 months. Secondary outcomes include health related quality of life, wellbeing, health care utilization, self-efficacy, health literacy, sick leave status (if applicable) and costs (each at 6-months-follow up and 12-months follow-up). Alongside the RCT, we will conduct a health economic evaluation of SP and related implementation activities. Analyses will assess economic aspects of SP and related implementation activities for each of the three target groups and in total (cost-effectiveness analysis).

[2] Hawkins, J., Madden, K., Fletcher, A. et al. Development of a framework for the co-production and prototyping of public health interventions. BMC Public Health 17, 689 (2017). doi:10.1186/s12889-017-4695-8

[3] Following the access model of Levesque, JF., Harris, M.F. & Russell, G. Patient-centred access to health care: conceptualising access at the interface of health systems and populations. Int J Equity Health 12, 18 (2013). doi: 10.1186/1475-9276-12-18 and the SP definition of Muhl C, Mulligan K, Bayoumi I, et al Establishing internationally accepted conceptual and operational definitions of social prescribing through expert consensus: a Delphi study. BMJ Open 2023;13:e070184. doi: 10.1136/bmjopen-2022-070184



We will also develop a **web-based open-source technical solution**, which will help link workers assemble, curate, and classify a list of local offerings by scraping public GIS sources. This will support link workers to select relevant offers tailored to the needs to the specific patients by using a deep-learning based language model that uses a textual description of the patient's needs. The AI component of the software solution will be technically robust following the seven requirements of trustworthy AI as outlined in the Assessment List for Trustworthy AI (ALTAI): *1. Human Agency and Oversight:* The software will only offer suggestions to link workers. Decisions are not taken by the software but by link worker and patient together. *2. Technical Robustness and Safety:* The software will run separately from the trial management being secure and resilient to attack. As an open source solution, it will be reliable and reproducible. *3. Privacy and Data Governance:* No personal data will be stored. The analysis will be conducted on a server located in the EU. *4. Transparency:* The software will be transparent as an open source solution. How the software works will be described in plain and easy language to link workers and patients. *5. Diversity, Non-discrimination and Fairness:* The software will be co-created with participants from persons in vulnerable situations to check for non-discrimination. *6. Societal and Environmental Well-being:* The software will support SP and thus have a positive environmental impact as well as a positive social impact. *7. Accountability:* We will monitor and report throughout the project any negative impacts of the software and document how these can be addressed and mitigated. We will conduct the ALTAI assessment during the development of the software to improve the trustworthiness.

In the **qualitative comparative part**, we are researching enabling and limiting factors to implementing SP by conducting qualitative interviews with stakeholders (e.g., patients, GPs, policy makers) in five different countries representing different stages of the implementation of SP and different forms of health care organization (i.e., Austria, Germany, Poland, Portugal, UK). We are going to conduct approximately 20 stakeholder interviews in each of the five countries and transcribe them verbatim. Framework analysis will be used to manage and interpret data. It was developed for qualitative research with relatively large samples and for undertaking analysis as a team. It will involve developing summaries (in a table-like chart) of what participants said in an interview, which can be shared with members of the research team. Researchers employed on the project will be responsible for data management of a specific country/countries for which they will code data. These researchers will come together for regular meetings to compare and discuss coding with each other. Codes from across the countries will be used by the research team to develop a thematic framework to chart data against. Six charts will be produced. They will summarise data based on the same thematic framework, to help with identifying patterns and peculiarities across countries. Charts will allow us to compare and contrast experiences and understanding of SP among the countries involved. The results of the qualitative part will then help to prepare for the exploitation of the RCT results into health care policy and practice. Policy cost-effectiveness (PCE) as a conceptual framework to evaluate economic aspects of the intervention and its implementation will be applied. In a **second phase of co-creation,** the results of the RCT and of the qualitative comparative part are going to be interpreted jointly by researchers and representatives of all three groups of people in vulnerable situations. These jointly interpreted results will then be used to inform health and care practitioners, policy makers and the public about the results and deduce specific recommendations how to implement SP for vulnerable groups in Europe.

Key facts of the RCT			
Study design	International, multicentre, pragmatic, individually randomized controlled open basket trial		
Population	Adults (18 years or older) capable of giving consent with at least one non-medical health related social need (ICPC-3: ZC01-ZC99 "Social Problems")		
	AND belonging to one of three vulnerable groups:	LGBTIQ	Persons who self-identify at least as one out of this list: homosexual, lesbian, gay, pansexual, bisexual, trans, inter, non-binary
		Refugees and first generation immigrants	Persons who live outside of their (non-Western) country of origin as a result of forced or voluntary migration
		Older adults living alone	Persons at least 65 years old and living in a single-person household
Intervention	Social prescribing (GP refers to a link worker, up to six consultations in up to three months with a link worker who connects the patient to local community services and offers); the intervention is adapted to each vulnerable group during the co-creation process (e.g., identification of vulnerability, asset mapping, specific training for link workers and GPs, tailoring to specific preferences)		
Comparison	Care as usual		
Randomization	Central block randomization with varying block size stratified by study center		
Primary Outcome	Utilization of non-clinical supports and services within the community (proportion of participants who accessed at least one service within 6 months)		
Secondary Outcomes	Health related quality of life, wellbeing, self-efficacy, health literacy, health care utilization, sick leave status and costs		
Setting	Primary care (GP practices)		
Time frame	Recruitment period: 16 months (PM 20-36) Intervention: 19 months (PM 20-39) Follow up: 6 months (primary endpoint), 12 months First patient in to last patient out: 28 months (PM 20-48)		
Organization	10 (academic) primary care research networks (study hubs) in 8 countries with around 10 GP practices each (study centres)		
Basket trial design	Due to their heterogeneity, the three vulnerable groups serve as participants of independent sub-trials. The management of the sub-trials is streamlined (project management, data management, study personnel) to maximize resources. Selection of study centres will ensure recruitment goals for all sub-trials.		
Sample size calculation	For each of the three sub-trials: Alpha two-sided 0.05, power 90%, chi-squared test, difference in proportions of utilisation (primary endpoint): 34% (control group) vs. 50% (intervention group). A sample size of 198 participants is needed per treatment group; to compensate for a dropout rate of about 33%, 296 participants per group are needed to be included in each sub-trial. For the entire project, 1,776 participants will be included.		
Analyses	1) Treatment policy estimand for the primary endpoint, individually for each sub-trial using a hierarchical mixed effects model including study hub and study centre (random effects) and treatment group (fixed effect); confirmatory analysis 2) Secondary endpoints: hierarchical mixed effects models 3) Subgroup analyses for factors, such as socio-economic and legal status, age, sex and gender identity, (dis)ability, ethnicity and geographical location 4) Pooling of data from the sub-trials in an individual participant data meta-analysis 5) Cost-effectiveness analysis The primary evaluation is performed with the full analysis set (FAS) based on the intention-to-treat principle (ITT) without replacing missing outcome values. Sensitivity analyses include also per-protocol (PP) analyses.		

Relation of SP-EU to the six climate and environmental objectives ("EU Taxonomy"): SP contributes to climate change mitigation as working and housing conditions as well as food security can be reasons for SP. So called "green" SP links people to nature-based activities, such as community gardening, mindful nature walks or volunteering in environmental organizations. There is evidence that SP results in the reduction of over-prescription of medications. This can help reduce carbon emissions as the manufacture and distribution of medicines, and the use of some medicines, has a significant impact on greenhouse gases. SP-EU does not do any significant harm, it does not harm

climate change adaption nor pollute water and marine resources; SP-EU does not influence the transition to a circular economy, influence pollution prevention and control nor harm biodiversity and ecosystems.

National or international research and innovation activities linked to SP-EU: Numerous national, European, and global networks dedicated to SP are actively engaged in advancing education, policy development, and research within this field. SP-EU seeks to leverage the valuable contributions made by these networks and actively involve them in the dissemination and utilization of our research outcomes.

Furthermore, there are several national RCTs focusing on SP, both ongoing and planned, within Europe and beyond. These trials address distinct target populations; for instance, an Irish RCT concentrates on primary care patients with polypharmacy. It is noteworthy that these trials are strictly confined to a single country. Consequently, our project can draw from the experiences gained in these trials while extending its reach far beyond national borders. Principal investigators from these trials are consulted to contribute their insights to SP-EU and serve as members of the external advisory board.

We intend to collaborate closely with other funded projects from this call to collectively harness our results for the betterment of healthcare access and services for people in vulnerable situations.

Inter-disciplinary approach: SP-EU presents a formidable challenge that necessitates the collaboration of a multidisciplinary team. The consortium comprises world-renowned experts, including family physicians, epidemiologists, social scientists, statisticians, public health specialists, and software developers, who collectively work towards realizing the overarching vision of SP-EU. These diverse fields of expertise are woven into every facet of SP-EU. Moreover, patients and citizens play an integral role in the entire project, expanding the interdisciplinary approach into a transdisciplinary one. To foster such collaboration, SP-EU can leverage established local, national, and international networks that bring together these experts and citizens.

Integration of social sciences and humanities: Social sciences and humanities (SSH) are an integral part of the methods and the consortium of SP-EU. In the consortium, researchers from a wide range of SSH background are represented: The co-creation process is led by researchers, e.g., with a background in social anthropology. The RCT is complemented by process and economic evaluations conducted by SSH researchers, which includes a health economist. The qualitative comparative part is led and undertaken by SSH researchers, e.g., with a background in sociology.

Gender Dimension: Sex and gender differences in health behaviour, health outcomes, health care experiences exist over the life course. Sex, gender, and an intersectional perspective are seamlessly integrated into SP-EU. This project will address sex and gender issues through following the Gendered Innovation Engineering checklist, which has been defined by the EC. Within the co-creation phase, it is of paramount importance to actively involve individuals of all sexes and genders, with a particular emphasis on including those who navigate intersectionality. In the RCT, we will meticulously examine variables such as gender, sexual orientation, migration history, and socio-economic status to assess how sex, gender, and intersectionality may impact the trial outcomes. These assessments will be conducted as both sensitivity analyses and subgroup analyses. In the qualitative comparative phase, we will purposively sample to achieve variation among participants in terms of gender, age, ethnicity and sexuality. Additionally, we will specifically inquire about the aspects of sex, gender, and intersectionality concerning the barriers and resources associated with the implementation of SP.

Implementation of open science practices: As researchers, we have the responsibility to adopt processes that increase the likelihood that research is trustworthy, useful for scientists and society, and ethical for humans. We will employ a wide array of open science practices to ensure that these goals are met. While there is little reason to suggest publication bias in the field of SP, reproducibility of and trustworthiness in the results have been suffering from missing open science practices, on which we will improve in SP-EU. The consortium is already actively involved in shaping open science practices, e.g., through research governance activities.

Early and open sharing of research: We will actively work towards a culture of early and open sharing of our research. To achieve this goal, we will prospectively register all parts of SP-EU, and publish protocols for all sub-trials, as well a Statistical Analysis Plan (SAP) for the sub-trials and meta-analysis. All results will be published as pre-prints to ensure timely access. All data sets will be made available before SP-EU ends.

Systematic research output management: To ensure easier access to valuable research resources and ensure the longevity of our scientific findings we will develop and implement strategies that facilitate the organization, documentation, and dissemination of research outputs, including data, software, and workflows. The software developed in the project will be published under open-source software licensing. This includes administration and support to ensure the sustainability of the software.

Measures to ensure reproducibility of research outputs: SP-EU will be supported by innovative and comprehensive institutional initiatives to improve open science practices. This includes training of all researchers involved in the project through local, national, and international workshops. We will also integrate a “spoke and hub” model with the goal of strengthening the rigor, reproducibility, and transparency in biomedical research beyond the

project. We will designate an "open science ambassador" who is specifically trained to provide resources to the rest of the research team to improve their knowledge, research methodologies, documentation, and data sharing practices.
Open access to research outputs: We will make all our research output accessible through immediate open access via a trusted repository under CC0. All information needed to validate or re-use the data will be provided as early as possible in the research process.
Early sharing of methods and participation in open peer-review: We will actively participate in open peer-review processes to foster transparency and accountability in the evaluation of research. In addition to favouring publication in journals with open identity and public reviews, we will focus our attention on open participation where we will invite all relevant knowledge actors in the review process.
Involving all relevant knowledge actors including citizens, civil society and end users in the co-creation of R&I agendas and contents (such as citizen science): In addition to the already described co-creation process, SP-EU will also contribute to this goal through the integration and cooperation of many relevant stakeholders and the translation of the initiatives into the broader community. We will produce tailored information for all relevant actors. This includes adaptation to the vulnerable groups under consideration, as well as translation to all languages that are represented in the multinational consortium. There are no legitimate interests of the consortium (e.g., commercialization) that hinder sharing of research data, except for data privacy reasons.
Research data management and management of other research outputs: We will ensure high-quality data management including the creation and recurrent updating of a joint data management plan (DMP). The DMP describes how we will handle data in accordance with the FAIR principles. We consider the DMP to be a "living document", which needs to be updated on a regular basis to incorporate current insights and best practices. SP-EU will adhere to the FAIR principles as published by GO FAIR (see below: F1-R1.3), as well as the Sorbonne declaration on research data rights:
Types of data/research outputs: We will make all relevant research outputs available. This includes data sets for the RCT (incl. code books and extensive metadata), data management and SAP, code for the processing and analysis of the data sets, pre-prints, publications, external presentations and reports (e.g., policy briefings).
Findability of data/research outputs: F1. (Meta) data are assigned a globally unique and persistent identifier. F2. Data are described with rich metadata (defined by R1 below). F3. Metadata clearly and explicitly include the identifier of the data it describes. F4. (Meta) data are registered or indexed in a searchable resource.
Accessibility of data/research outputs: A1. (Meta) data are retrievable by their identifier using a standardized communications protocol. A1.1 The protocol is open, free, and universally implementable. A1.2 The protocol allows for an authentication and authorization procedure, where necessary. A2. Metadata are accessible, even when the data are no longer available.
Interoperability of data/research outputs: I1. (Meta) data use a formal, accessible, shared, and broadly applicable language for knowledge representation. I2. (Meta) data use vocabularies that follow FAIR principles. I3. (Meta) data include qualified references to other (Meta) data.
Reusability of data/research outputs: R1. (Meta) data are richly described with a plurality of accurate and relevant attributes. R1.1. (Meta) data are released with a clear and accessible data usage license. R1.2. (Meta) data are associated with detailed provenance. R1.3. (Meta) data meet domain-relevant community standards. As evidence by F-UJI shows that R1 and R1.2 are currently underutilized, we will focus on these aspects.
Curation and storage/preservation costs: We will utilize the general-purpose repository for research data objects Zenodo. This repository is developed under the European OpenAIRE program and is funded by the European Commission. It is free-of-charge for users and there is a low risk for termination of the project. It provides features to adhere to all FAIR principles mentioned above and currently achieves the highest FAIRness score by F-UJI.
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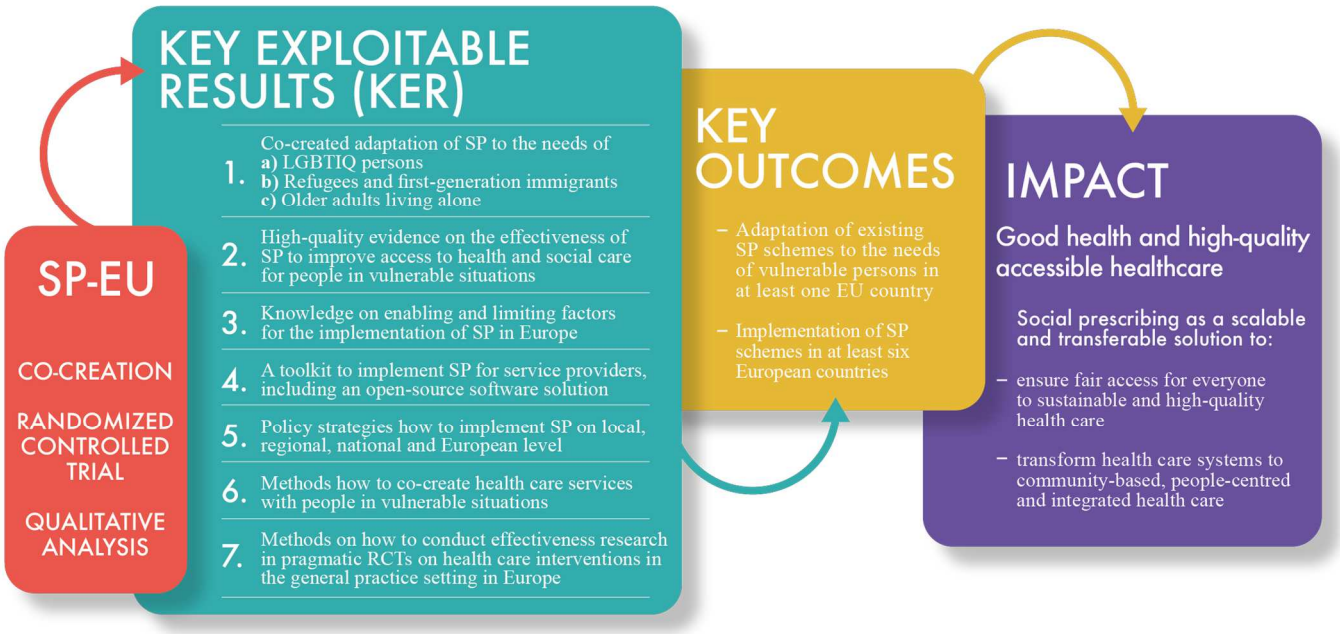
2 Impact #@IMP-ACT-IA@#

2.1 Project's pathways towards impact

Contribution to sustainable development

Contribution of project results towards the expected outcomes and wider impacts

SP-EU will contribute to the goal **creating a more resilient, inclusive and democratic European society** of the Horizon Europe Strategic Plan 2021-2024 by **addressing inequalities and providing high-quality health and social care**. Thus, SP-EU contributes to the impact area "**Good health and high-quality accessible healthcare**".
Regarding the expected impacts in the area of **Ensuring access to innovative, sustainable and high-quality health care** of **cluster health**, we will contribute especially to the following pathway to impact:



Given our ambition to make SP implementable across as many regions of the European Union as possible, we will need to engage a wide range of relevant stakeholders with targeted messages, information and tools. These tools will empower decision-makers and stakeholders, including active citizens, to fully understand and implement the project outputs and deliverables relevant to and optimized for local benefits. The initial exploitation plan will be developed by PM 36 of the project and will be based on our main key exploitable results (KERs). The KERs will be chosen together with stakeholders regarding their potential for exploitability and impact. We have chosen seven provisional KERs because they have the highest potential to be exploited for policy and research at the end of SP-EU (cf. figure).
KER1 (relates to Obj.1, KPI 1.1) Exploitation targeting decision- and policymakers, services providers, health and care workers and patients by policy recommendations, manuals and social media activities
KER2 (relates to Obj.2, KPI 2.1-2.7) Exploitation targeting decision- and policymakers, services providers, health and care workers by different measures such as scientific publications, conference presentations, manuals, online educational sources, workshops and policy roundtables
KER3 (relates to Obj.3, KPI 3.1-3.2) Exploitation targeting decision- and policymakers on local, regional, national and European level by manuals, workshops and policy roundtables
KER4 (relates to Obj.1&4, KPI 1.2, 4.1-4.3, 5.2) Exploitation by making the toolkit and accompanying educational resources available via an open source webpage
KER5 (relates to Obj.4, KPI 2.6, 3.2 4.1-4.3) Exploitation targeting decision- and policymakers by manuals, workshops and policy roundtables
KER6 (relates to Obj.1, KPI 1.3) Exploitation targeting decision- and policymakers, service providers, as well as researchers with a manual and accompanying educational resources
KER7 (relates to Obj. 2, KPI 2.1-2.7) Exploitation targeting scholars, researchers and practitioners by research articles, conference workshops, information material and social media.

These KERs will contribute to two **major mid-term outcomes**:
Key Outcome 1: In at least one EU country with an already existing SP scheme, the scheme will be adapted to the needs of persons in vulnerable situations (based on KER1, KER2, KER4, KER6).
Key Outcome 2: In at least six EU countries, SP schemes have been implemented at least on a regional level to enhance access for persons in vulnerable situations (based on KER2, KER3, KER4, KER5).

The following table shows, how KERs and outcomes contribute to the **expected outcomes described in the call**:

Expected Outcome from the Call	How SP-EU contributes
Decision- and policymakers, service providers, and health and care workers have better availability to and make use of knowledge on barriers to access to health and care by people in vulnerable situations and at risk of stigma or discrimination.	The knowledge of barriers ascertained in the co-creation process and the qualitative part of SP-EU will be spread to decision and policymakers via policy briefing and specific workshops as well as manuals for local decision makers and to health and care workers via courses building up on the knowledge created in SP-EU as well as manuals for health care workers. These include actions to address barriers via SP.

Decision- and policymakers, providers and health and care workers have access to innovative solutions to promote and improve access to health and care services for people in vulnerable situations.	SP as an innovative solution will be ready to be implemented across Europe by the end of SP-EU. Manuals, courses, workshops, and roundtables targeting different levels of decision- and policymakers, providers and health and care workers enable them to implement SP to promote and improve access to health and care services for people in vulnerable situations.
People in vulnerable situations are better equipped in terms of health and digital literacy, knowledge about their rights etc. when it comes to access to health and care services.	SP empowers people in vulnerable situation. The results of SP-EU will include manuals, workshops and courses to facilitate the implementation of SP, which then will equip persons in vulnerable situations with necessary tools to access relevant health and social services.
People in vulnerable situations are involved in the design and implementation of research and innovation activities concerning access to health and care services.	With the co-creation process in SP-EU, we demonstrate how people in vulnerable situations can be meaningfully involved in the design and implementation of research and innovation activities. We will publish and disseminate a manual describing the approach and enabling scholars in Europe to involve people in vulnerable situations in research and innovation activities.

Building on these expected outcomes, we describe in the following, how we are going to achieve the **expected impact:**

Health care systems provide equal access to innovative, sustainable and high-quality health care thanks to the development and uptake of safe, cost-effective and people-centred solutions, with a focus on population health, health systems resilience, as well as improved evidence-based health policies

SP - especially in our adapted version (KER1) - is a people-centred solution. We will show the effectiveness of SP including the cost-effectiveness (KER2). With the knowledge on enabling and limiting factors to implement SP (KER3), the toolkit to implement SP (KER4) and policy strategies (KER5), we will prepare the ground for a sustainable uptake of SP via evidence-based health policies. We will make these KERs exploitable by different measures. A key measure will be manuals tailored to the needs of decision- and policymakers, service providers, and health and care workers to implement SP on different levels (e.g., local, regional, national). These will include a flexible framework on financial sustainability of SP. Explanatory materials, such as social media activities, films and other online resources, will accompany these manuals. Educational material will allow the uptake of SP in daily practice. Additionally, we will promote our results via policy roundtables and policy briefings, communicating with politicians and policy makers on European and national level. By these measures, we will prepare the ground for a stepwise uptake of SP schemes in an increasing number of EU countries (Key Outcome 2).

As there are also existing SP schemes, our key results (especially KER1 and KER4, but also KER6) will allow adjusting these existing schemes to the needs of persons in vulnerable situations. Thus, SP-EU will contribute to enhance the positive impacts of these existing schemes even further (Key Outcome 1).

Both together will strengthen European health care systems to be more sustainable with an improved access to health and social care. SP will improve the amplification of individual primary care with a population health perspective.

At a more general level, the outcomes of SP-EU will pave the way for the development and adoption of innovative and sustainable healthcare solutions through KER6 and KER7. KER6 will empower stakeholders in the healthcare sector to apply the knowledge acquired during our co-creation process to design healthcare interventions for people in vulnerable situations. This knowledge will be disseminated through a comprehensive manual accompanied by supplementary educational materials and conveyed through workshops targeting knowledge multipliers. KER7 will serve as an exemplary guide on how to conduct rigorous and pragmatic international multicentre trials in primary care, particularly focusing on healthcare interventions. Our experiences will serve as a blueprint for others to follow, leveraging the established networks of study hubs, researchers, and practitioners. Consequently, this will establish a foundation for future European primary care healthcare interventions, akin to the groundwork laid by ALIC4E in preparing the landscape for initiatives such as PREPARE and PANORAMIC, which address infectious diseases in primary care.

Shifting from hospital-centred to community-based, people-centred and integrated health care structures

SP-EU will contribute to a broader transformation of health care in Europe. SP widely established in primary care in Europe will be people-centred due to the co-creative nature of the SP-EU adaptation. SP will strengthen the community-orientation of primary care by fostering connections between GPs and community resources. We will ensure this impact by disseminating our results both to practitioners and to communities including care providers in the community as well as service providers. The uptake of SP on a European level will contribute to the integration of health, care and social structures at national levels. Our results pertaining to implementation and policy will facilitate the incorporation of SP into integrated healthcare structures.

In summary, SP, as further developed by SP-EU, will be a powerful tool in the toolkit to transform healthcare towards community-based, people-centred and integrated health, care and social structures.

Scale and significance of the project's contribution to the expected outcomes and impacts

To show the scale and significance of SP-EU, we focus on **loneliness as example** for a well-researched and important social problem, which can be tackled by SP. The base of our estimation are population figures from Eurostat, which estimates that **up to 98 million persons in the EU are members of the three vulnerable groups** (cf. the following table for detailed numbers). From existing epidemiological studies, we estimated the number of lonely persons in each group. Foster et al. showed a significant reduction of loneliness through SP by 16 percentage points more than controls[4]. Considering that loneliness decreases healthy life years by 4.8 years per person[5], we come up to a potential of SP-EU to improve the healthy life expectancy by 17.6 million years if implemented all over the EU and all patients take it up. However, we assume a 10% uptake as realistic, leading to an **increase of healthy life expectancy in the EU by 1.76 million years**.

Vulnerable Group	Estimated population in EU (in Million)	Prevalence of loneliness (in Million)	Of these potentially benefitting from SP by not being lonely anymore (in Million)	Resulting potentially improved health life years (in Million)
LGBTIQ	26.5	9.4 (35.5%[6])	1.5	7.2
Refugees and 1st generation immigrants	38.0	8.4 (22.1%[7])	1.3	6.2
Older adults living alone	29.5	5.3 (18.0%[8])	0.8	4.2
Total:	94.0	23.1	3.6	17.6

Regarding the **Social Return on Investment (SROI)**, Kimberlee et al.[9] gave an overview of conducted studies. The social return on investment varied in these studies between **1.9 and 4.9 Euros social return per invested Euro**. Thus, SP will be cost-effective and the implementation of SP on a European level will bring a strong social return.

Requirements and potential barriers:

Potential Barrier	How we address the barrier
Workforce considerations: There is a workforce shortage especially in the health and social care sector. Finding no link workers to implement SP might be a barrier to implementing SP broadly in the EU.	The SP concept of SP-EU will free GPs from the burden of additional tasks with social problems. Thus, it will soften the workforce shortage of GPs. The UK already has experience recruiting and training link workers. We will learn from these experiences through the qualitative part of SP-EU. Training manuals from the RCT can be used to train link workers after SP-EU. The lower training requirements for link workers compared to nurses and social workers will ease the workforce shortage in case of SP.
Public acceptance: The public might not be willing to accept a referral from primary care to community services.	Experiences from the UK show a high acceptance of SP. However, we will address this barrier in the qualitative part of SP-EU and by having a good communication strategy involving social media and classic mass media in communicating with the public.
Financing: To implement SP, there will be two financial challenges. First financing the link worker and the process of SP and second financing the community services.	We will address the issue of financing community services in the qualitative part of the study. There are going to be different local solutions in each of the EU countries to cover these problems, as the community and NGO sectors are differently financed throughout the EU. With the data of the economic analysis from the RCT, we are going to be able to prove the financial sustainability of SP. We will develop a framework for sustainable financing of SP in European countries.

#§IMP-ACT-IA§#

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